In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt
```

```
C:\Users\neeti\AppData\Roaming\Python\Python313\site-packages\pandas\core\computation\expressions.py:22: UserWarning: Pandas re
quires version '2.10.2' or newer of 'numexpr' (version '2.10.1' currently installed).
  from pandas.core.computation.check import NUMEXPR_INSTALLED
```

In [2]:
```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
```

In [3]:
```
df = pd.read_csv("N:\Download\project\Sample - Superstore.csv", encoding='latin1')
```

In [4]:
```
print(df.info())
```

```
<class 'pandas.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Row ID         9994 non-null   int64
 1   Order ID       9994 non-null   str
 2   Order Date     9994 non-null   str
 3   Ship Date      9994 non-null   str
 4   Ship Mode      9994 non-null   str
 5   Customer ID    9994 non-null   str
 6   Customer Name  9994 non-null   str
 7   Segment        9994 non-null   str
 8   Country        9994 non-null   str
 9   City           9994 non-null   str
 10  State          9994 non-null   str
 11  Postal Code    9994 non-null   int64
 12  Region         9994 non-null   str
 13  Product ID     9994 non-null   str
 14  Category       9994 non-null   str
 15  Sub-Category   9994 non-null   str
 16  Product Name   9994 non-null   str
 17  Sales          9994 non-null   float64
 18  Quantity       9994 non-null   int64
 19  Discount       9994 non-null   float64
 20  Profit         9994 non-null   float64
dtypes: float64(3), int64(3), str(15)
memory usage: 3.3 MB
None
```

In [5]:
```
print(df.head())
```

```
   Row ID        Order ID  Order Date   Ship Date       Ship Mode Customer ID  \
0       1  CA-2016-152156   11/8/2016  11/11/2016    Second Class    CG-12520
1       2  CA-2016-152156   11/8/2016  11/11/2016    Second Class    CG-12520
2       3  CA-2016-138688   6/12/2016   6/16/2016    Second Class    DV-13045
3       4  US-2015-108966  10/11/2015  10/18/2015  Standard Class    SO-20335
4       5  US-2015-108966  10/11/2015  10/18/2015  Standard Class    SO-20335

     Customer Name    Segment        Country             City  ...  \
0      Claire Gute   Consumer  United States        Henderson  ...
1      Claire Gute   Consumer  United States        Henderson  ...
2  Darrin Van Huff  Corporate  United States      Los Angeles  ...
3   Sean O'Donnell   Consumer  United States  Fort Lauderdale  ...
4   Sean O'Donnell   Consumer  United States  Fort Lauderdale  ...

  Postal Code  Region       Product ID         Category Sub-Category  \
0       42420   South  FUR-BO-10001798        Furniture    Bookcases
1       42420   South  FUR-CH-10000454        Furniture       Chairs
2       90036    West  OFF-LA-10000240  Office Supplies       Labels
3       33311   South  FUR-TA-10000577        Furniture       Tables
4       33311   South  OFF-ST-10000760  Office Supplies      Storage

                                       Product Name     Sales  Quantity  \
0                  Bush Somerset Collection Bookcase  261.9600         2
1  Hon Deluxe Fabric Upholstered Stacking Chairs,...  731.9400         3
2  Self-Adhesive Address Labels for Typewriters b...   14.6200         2
3      Bretford CR4500 Series Slim Rectangular Table  957.5775         5
4                     Eldon Fold 'N Roll Cart System   22.3680         2

   Discount    Profit
0      0.00   41.9136
1      0.00  219.5820
2      0.00    6.8714
3      0.45 -383.0310
4      0.20    2.5164

[5 rows x 21 columns]
```

In [6]:
```
print(df.isnull().sum())
```

```
Row ID           0
Order ID         0
Order Date       0
Ship Date        0
Ship Mode        0
Customer ID      0
Customer Name    0
Segment          0
Country          0
City             0
State            0
Postal Code      0
Region           0
Product ID       0
Category         0
Sub-Category     0
Product Name     0
Sales            0
Quantity         0
Discount         0
Profit           0
dtype: int64
```

In [7]:
```
df.isnull().sum().sum()
```

Out[7]:
```
np.int64(0)
```

In [8]:
```
df.drop_duplicates(inplace=True)
```

In [9]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Row ID         9994 non-null   int64
 1   Order ID       9994 non-null   str
 2   Order Date     9994 non-null   str
 3   Ship Date      9994 non-null   str
 4   Ship Mode      9994 non-null   str
 5   Customer ID    9994 non-null   str
 6   Customer Name  9994 non-null   str
 7   Segment        9994 non-null   str
 8   Country        9994 non-null   str
 9   City           9994 non-null   str
 10  State          9994 non-null   str
 11  Postal Code    9994 non-null   int64
 12  Region         9994 non-null   str
 13  Product ID     9994 non-null   str
 14  Category       9994 non-null   str
 15  Sub-Category   9994 non-null   str
 16  Product Name   9994 non-null   str
 17  Sales          9994 non-null   float64
 18  Quantity       9994 non-null   int64
 19  Discount       9994 non-null   float64
 20  Profit         9994 non-null   float64
dtypes: float64(3), int64(3), str(15)
memory usage: 3.3 MB
```

In [10]:
```
df.shape
```

Out[10]:
```
(9994, 21)
```

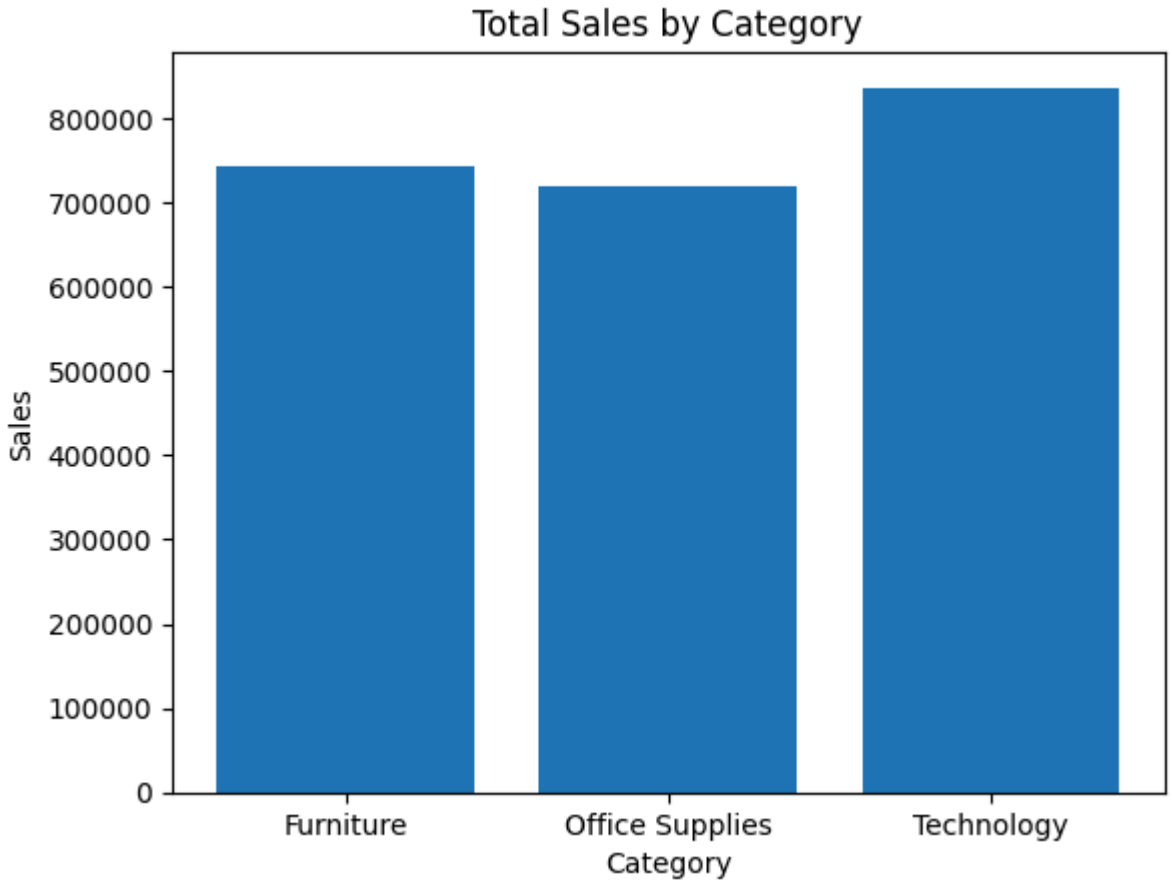
In [11]:
```
df = df.drop(['Row ID', 'Order ID', 'Customer ID', 'Customer Name'], axis=1)
```

In [12]:
```
df = df.drop_duplicates()
```

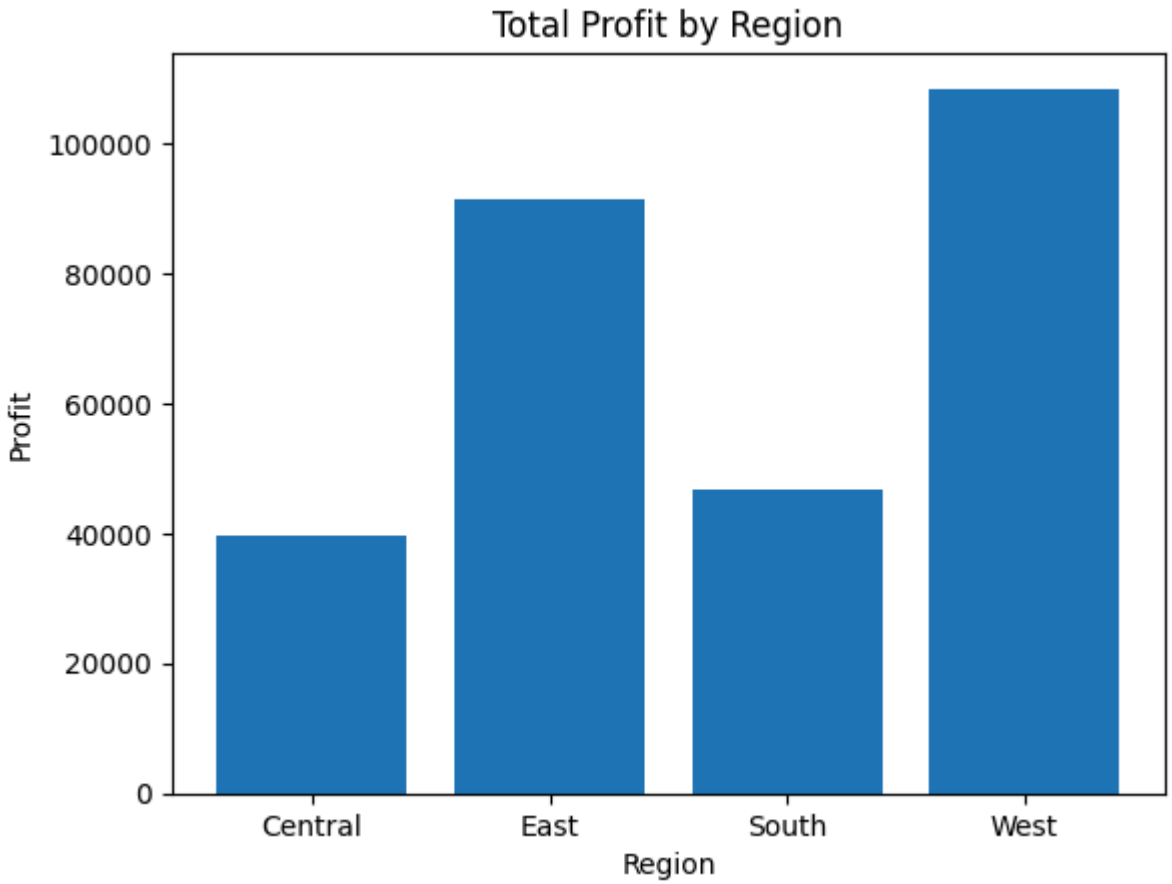
In [13]:
```
df = df.dropna()
```

In [14]:
```
df['Is_Profit'] = (df['Profit'] > 0).astype(int)
df['Target'] = df['Is_Profit']
```

In [15]:
```
sales_category = df.groupby('Category')['Sales'].sum()

plt.figure()
plt.bar(sales_category.index, sales_category.values)
plt.title("Total Sales by Category")
plt.xlabel("Category")
plt.ylabel("Sales")
plt.show()
```

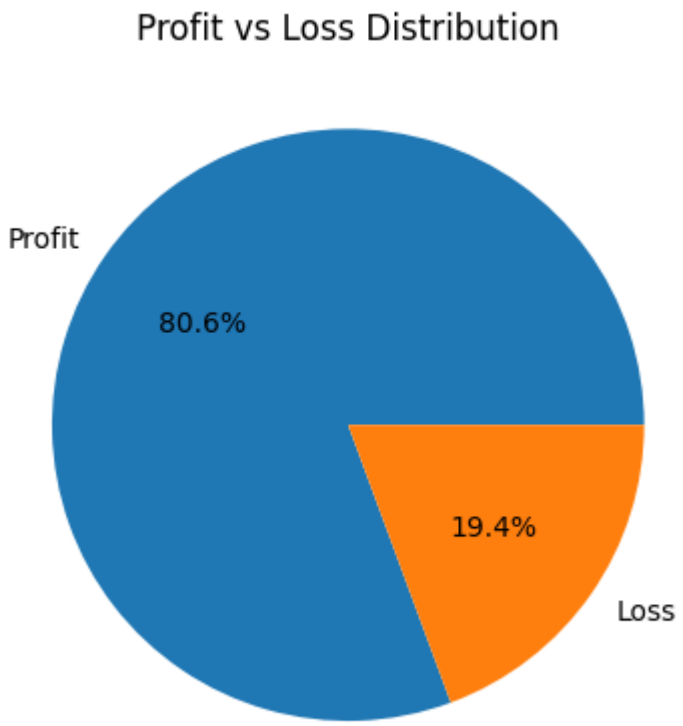


In [16]:
```
region_profit = df.groupby('Region')['Profit'].sum()

plt.figure()
plt.bar(region_profit.index, region_profit.values)
plt.title("Total Profit by Region")
plt.xlabel("Region")
plt.ylabel("Profit")
plt.show()
```

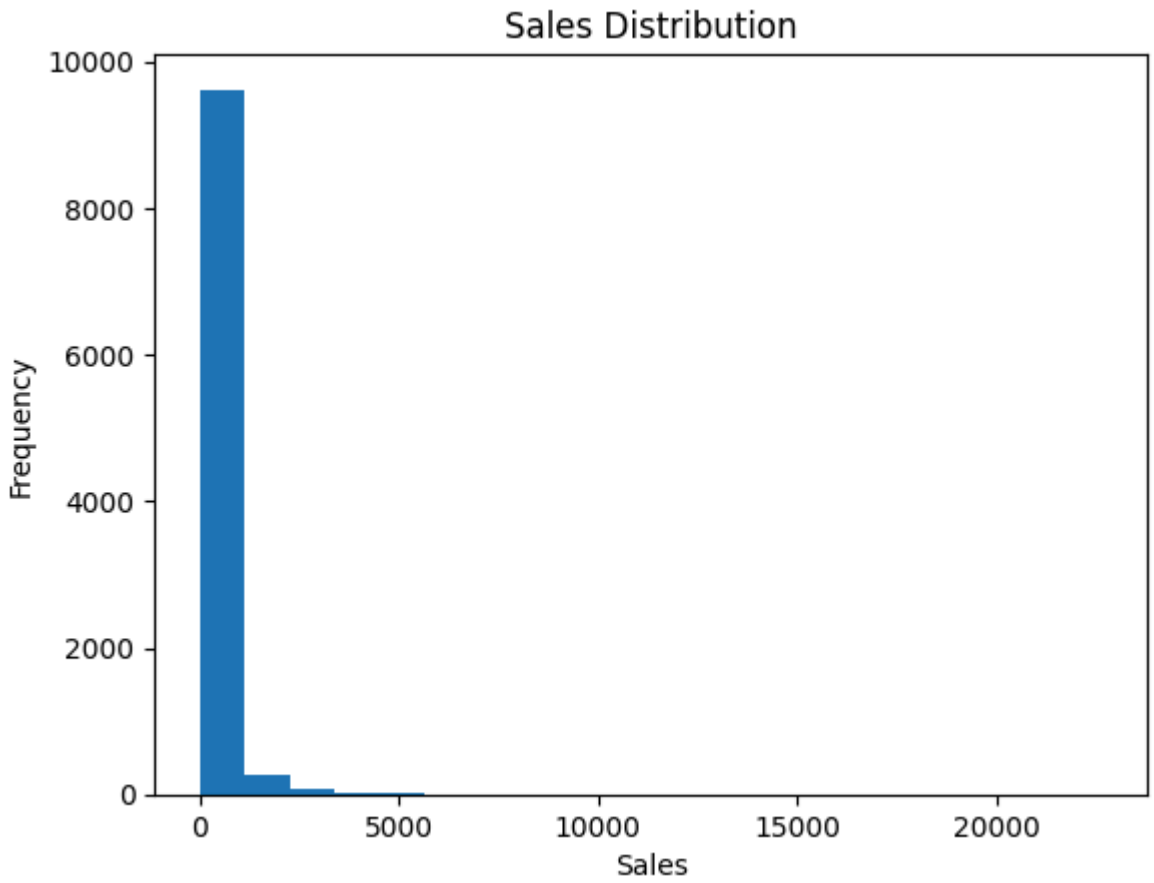


In [17]:
```
profit_count = df['Is_Profit'].value_counts()

plt.figure()
```

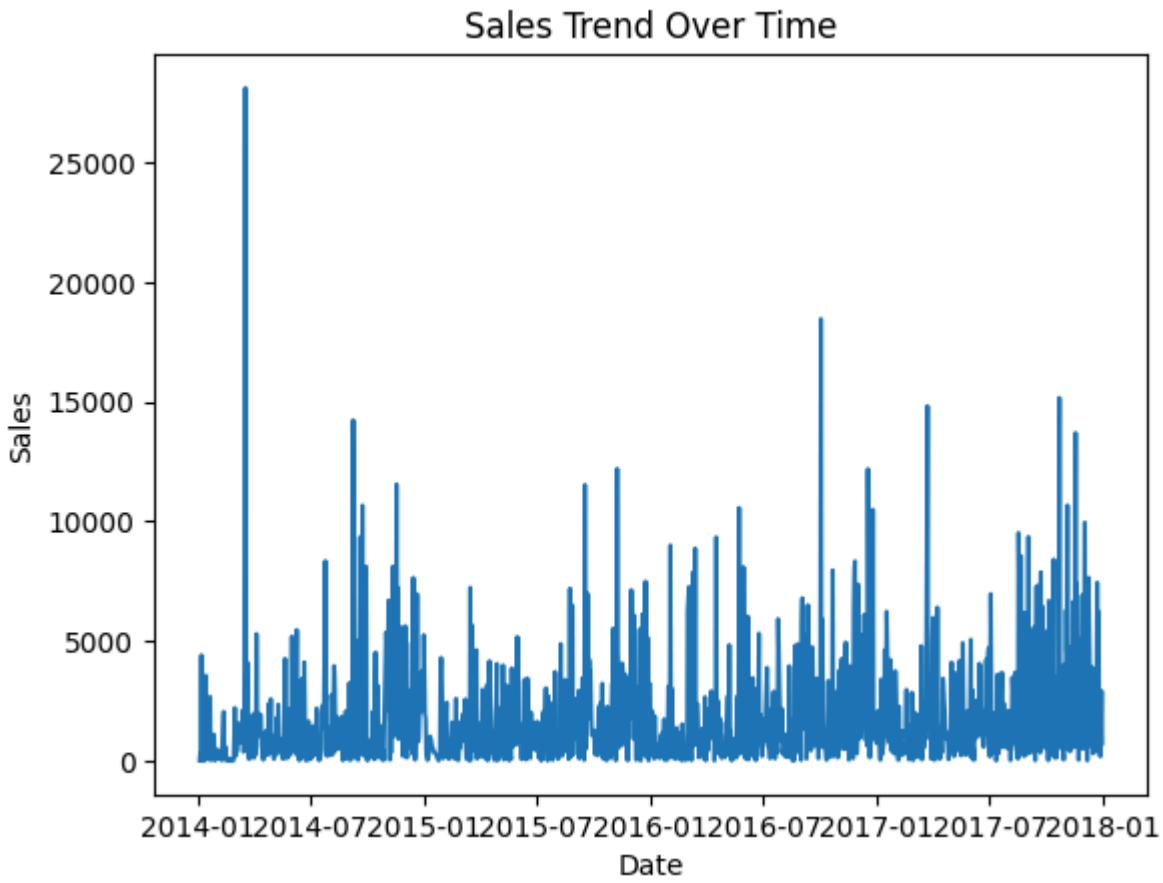
```
plt.pie(profit_count, labels=['Profit', 'Loss'], autopct='%1.1f%%')
plt.title("Profit vs Loss Distribution")
plt.show()
```



In [18]:
```
plt.figure()
plt.hist(df['Sales'], bins=20)
plt.title("Sales Distribution")
plt.xlabel("Sales")
plt.ylabel("Frequency")
plt.show()
```



In [19]:
```
df['Order Date'] = pd.to_datetime(df['Order Date'])

sales_trend = df.groupby('Order Date')['Sales'].sum()

plt.figure()
plt.plot(sales_trend.index, sales_trend.values)
plt.title("Sales Trend Over Time")
plt.xlabel("Date")
plt.ylabel("Sales")
plt.show()
```

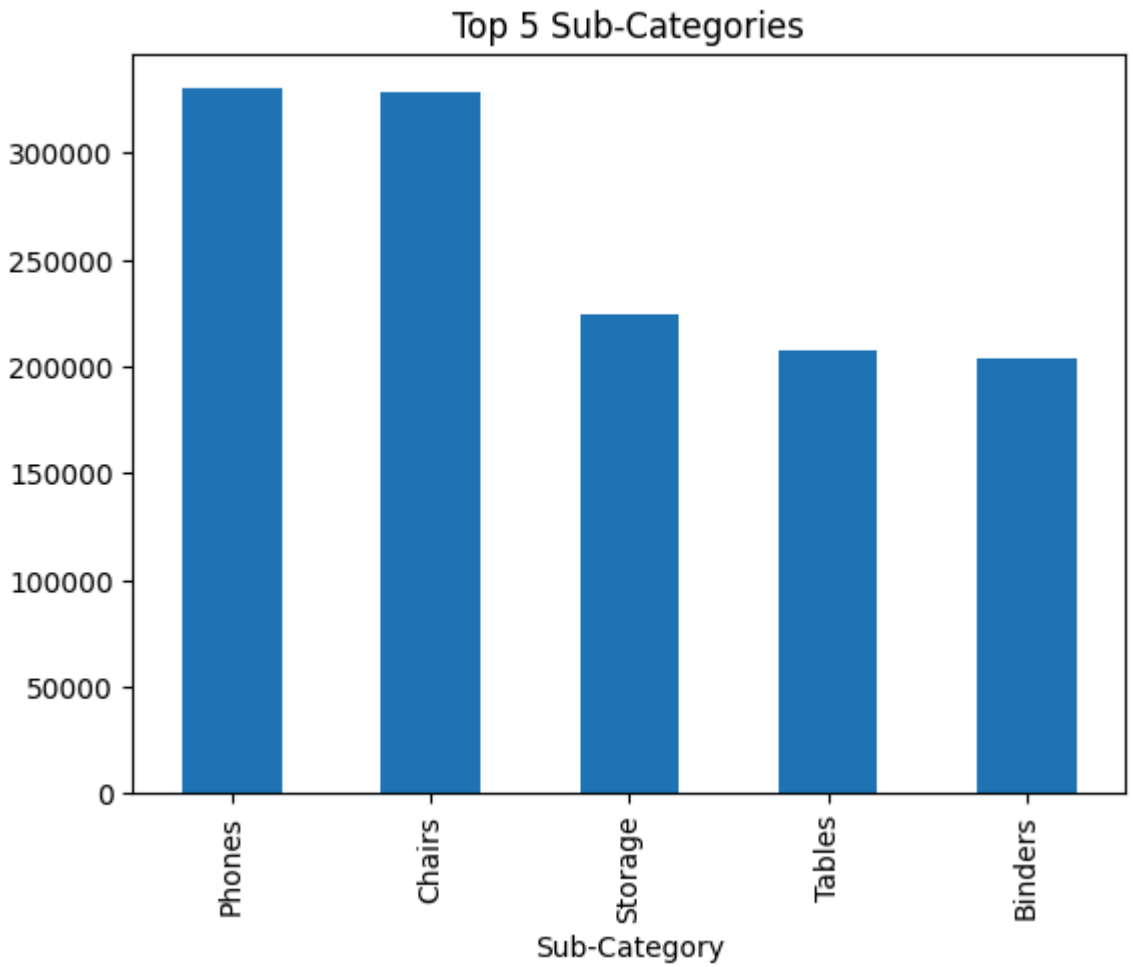


In [20]:
```
print(df.groupby('Category')['Sales'].sum())
print(df.groupby('Region')['Profit'].mean())
print(df.groupby(['Category','Region'])['Sales'].sum())
```

```
Category
Furniture          741718.4233
Office Supplies    719047.0320
Technology         836154.0330
Name: Sales, dtype: float64
Region
Central    17.092709
East       32.151331
South      28.857673
West       33.849032
Name: Profit, dtype: float64
Category         Region
Furniture        Central    163797.1638
                 East       208009.8320
                 South      117298.6840
                 West       252612.7435
Office Supplies  Central    167026.4150
                 East       205516.0550
                 South      125651.3130
                 West       220853.2490
Technology       Central    170416.3120
                 East       264973.9810
                 South      148771.9080
                 West       251991.8320
Name: Sales, dtype: float64
```

In [21]:
```
df['Category_Sales'] = df.groupby('Category')['Sales'].transform('sum')
df['Region_Profit'] = df.groupby('Region')['Profit'].transform('mean')
```

In [22]:
```
top_products = df.groupby('Sub-Category')['Sales'].sum().sort_values(ascending=False).head()

top_products.plot(kind='bar')
plt.title("Top 5 Sub-Categories")
plt.show()
```



In [23]:
```
category_sales = df.groupby('Category')['Sales'].sum()

percentage = (category_sales / category_sales.sum()) * 100
print(percentage)
```

```
Category
Furniture          32.291877
Office Supplies    31.304843
Technology         36.403280
Name: Sales, dtype: float64
```

In [24]:
```
df['Is_Profit'] = (df['Profit'] > 0).astype(int)
df['Target'] = df['Is_Profit']
```

In [25]:
```
df['Order Date'] = pd.to_datetime(df['Order Date'])
df['Ship Date'] = pd.to_datetime(df['Ship Date'])

# Extract features
df['Order Day'] = df['Order Date'].dt.day
df['Order Month'] = df['Order Date'].dt.month

df['Ship Day'] = df['Ship Date'].dt.day
df['Ship Month'] = df['Ship Date'].dt.month

# Drop original columns
df = df.drop(['Order Date', 'Ship Date'], axis=1)
```

In [26]:
```
from sklearn.preprocessing import LabelEncoder

for col in df.select_dtypes(include=['object','string']).columns:
    le = LabelEncoder()
    df[col] = le.fit_transform(df[col])
```

In [27]:
```
X = df.drop(['Profit', 'Is_Profit', 'Target'], axis=1)
y = df['Target']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
model = DecisionTreeClassifier()
model.fit(X_train,y_train)
```

Out[27]:
```
▾ DecisionTreeClassifier
▸ Parameters
```

In [28]:
```
y_pred = model.predict(X_test)
```

In [29]:
```
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
Accuracy: 0.9414707353676839
```

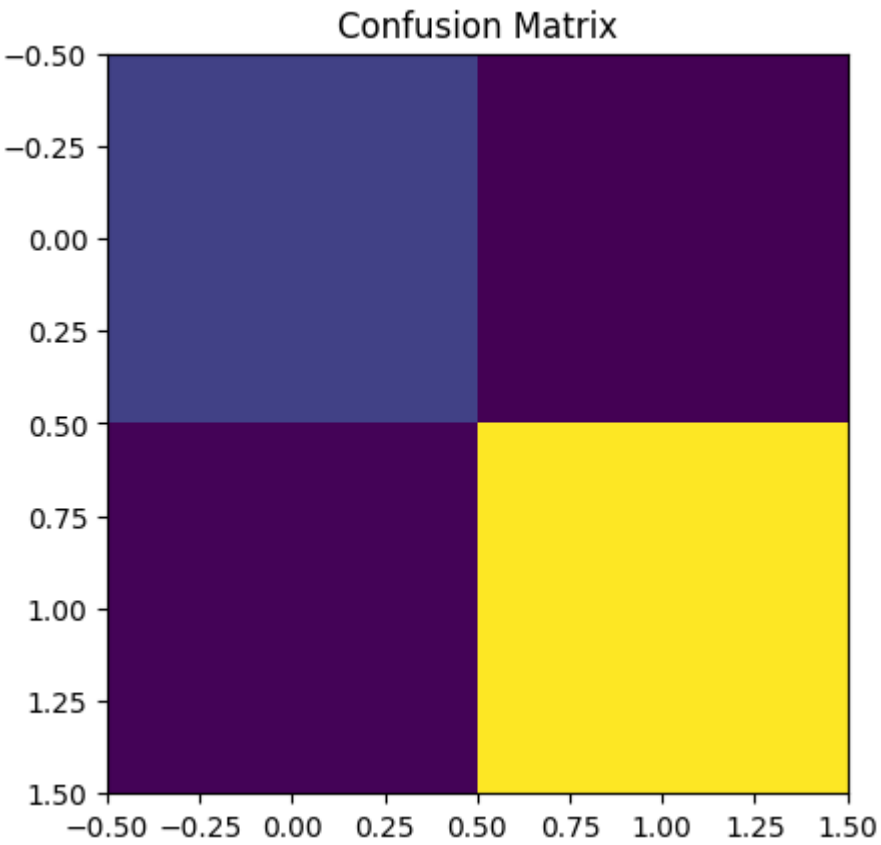
In [30]:
```
cm = confusion_matrix(y_test, y_pred)
print(cm)
```

```
[[ 338   50]
 [  67 1544]]
```

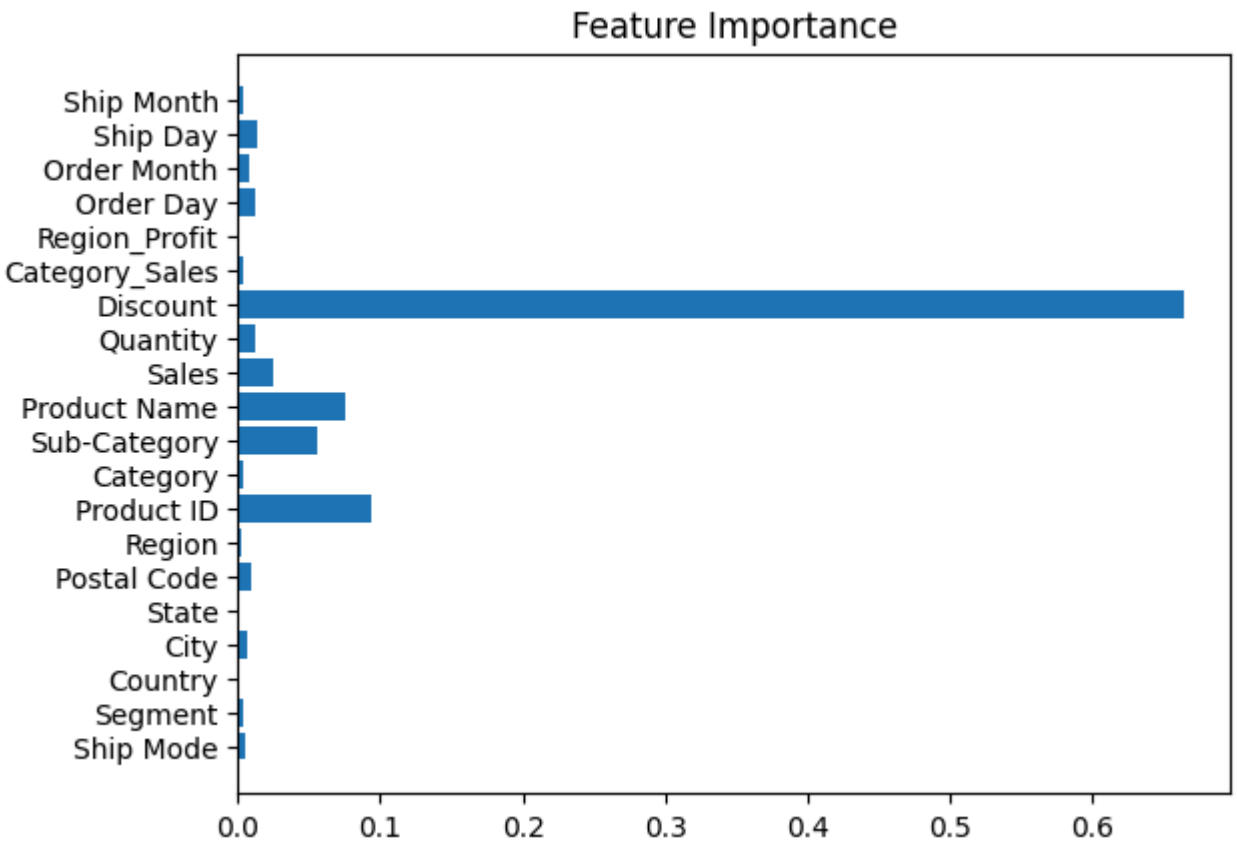
In [31]:
```
print(classification_report(y_test, y_pred))
```

```
              precision    recall  f1-score   support

           0       0.83      0.87      0.85       388
           1       0.97      0.96      0.96      1611

    accuracy                           0.94      1999
   macro avg       0.90      0.91      0.91      1999
weighted avg       0.94      0.94      0.94      1999
```

In [32]:
```
plt.imshow(cm)
plt.title("Confusion Matrix")
plt.show()
```



In [33]:
```
plt.barh(X.columns, model.feature_importances_)
plt.title("Feature Importance")
plt.show()
```


Feature Importance
Ship Month
Ship Day
Order Month
Order Day
Region_Profit
Category_Sales
Discount
Quantity
Sales
Product Name
Sub-Category
Category
Product ID
Region
Postal Code
State
City
Country
Segment
Ship Mode
0.0
0.1
0.2
0.3
0.4
0.5
0.6

In [ ]: